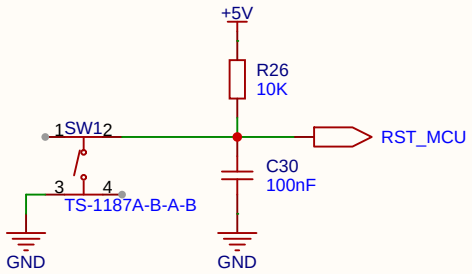
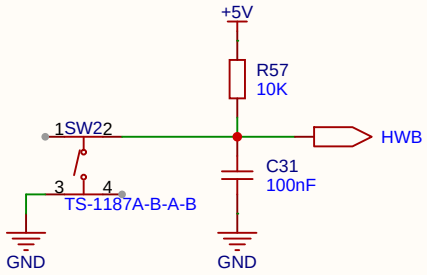
Fuse	Wert	Bedeutung
Low	0xFF	externer 16-MHz-Quarz, kein Taktteiler (wie Leonardo)
High	0xD1	BOOTRST aus, Bootsektion 4 KB, JTAG aus, EESAVE an
Extended	0xF0	HWBE an, BOD 4,3 V

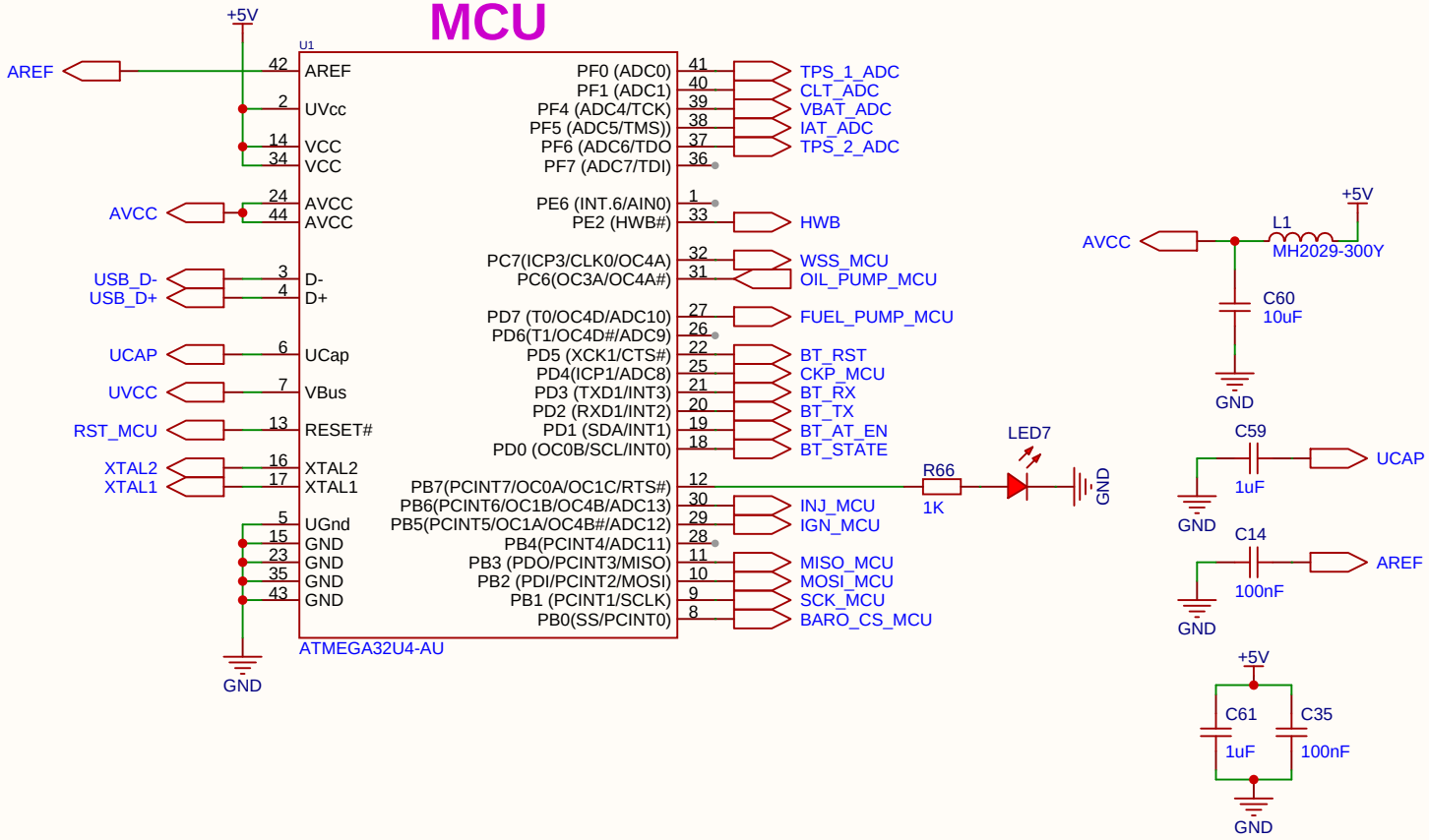
RESET



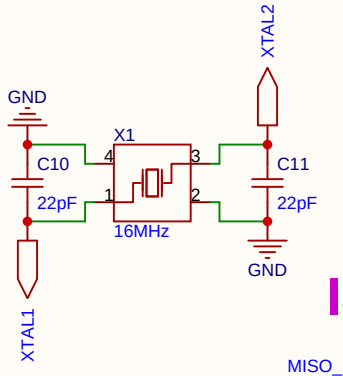
BOOT



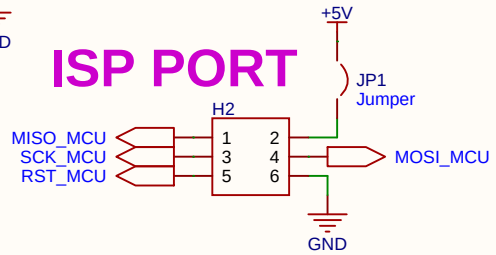
MCU



CRYSTAL 16Mhz

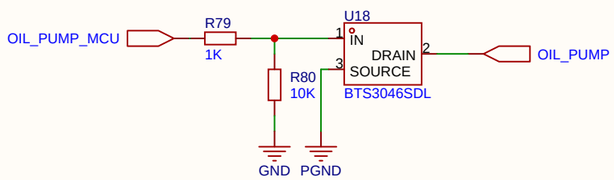


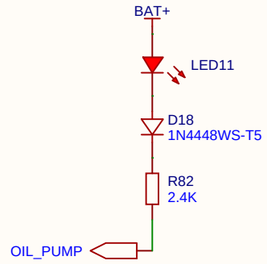
ISP PORT



Schematic	ZTech ECU v0.1	Create at	2026-10-30
		Update at	2026-11-02
Board	Board1	Page	MCU
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 16	
		EasyEDA.com	

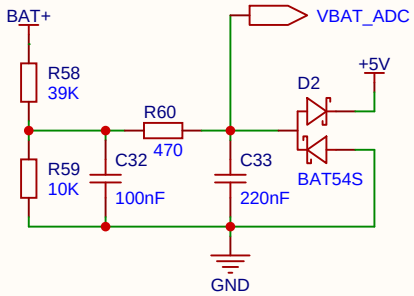
Oil Pump Driver





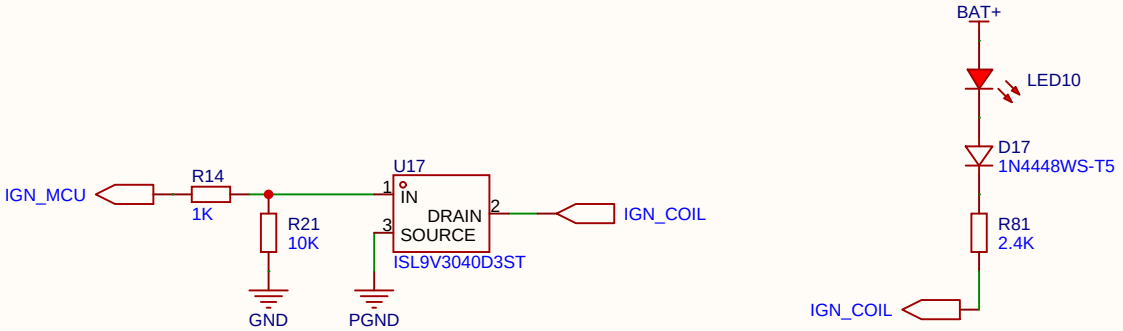
Schematic	ZTech ECU v0.1	Create at	2026-11-01
		Update at	2026-11-01
Board	Board1	Page	Oil Pump
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 2 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

Battery Reference Voltage



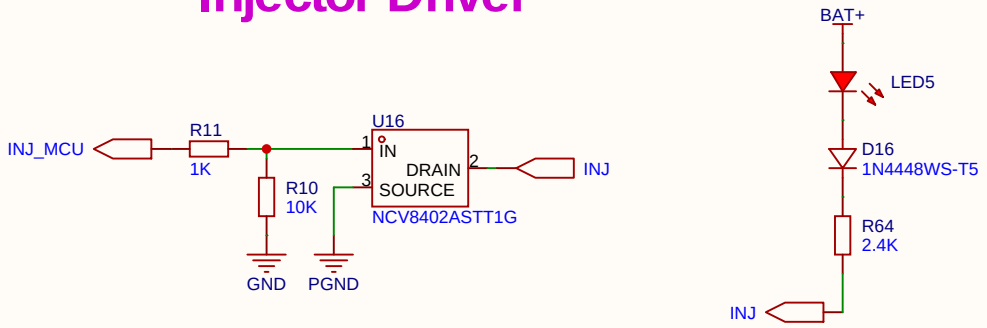
Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	Battery Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version / Size / Page 3 Total 16	
EasyEDA		V1.0 / A4	EasyEDA.com

Ignition Coil Driver



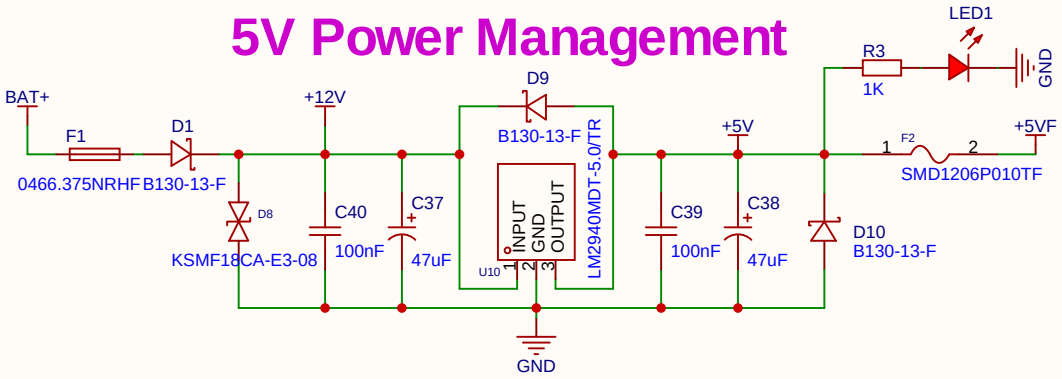
Schematic	ZTech ECU v0.1	Create at	2026-10-30
		Update at	2026-11-01
Board	Board1	Page	Ignition
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 4 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

Injector Driver

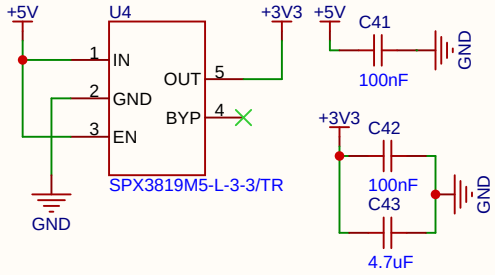


Schematic	ZTech ECU v0.1	Create at	2026-10-30
		Update at	2026-11-01
Board	Board1	Page	Injection
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version / Size / Page 5 Total 16	
EasyEDA		V1.0 / A4	EasyEDA.com

5V Power Management

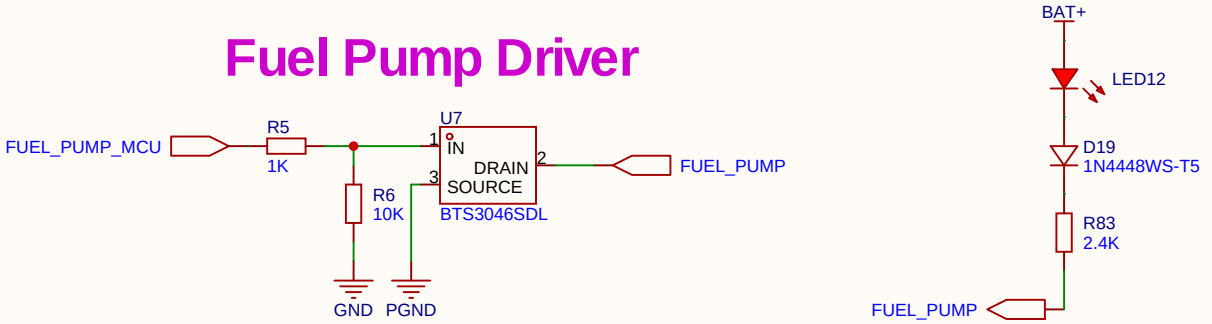


3.3V Power Management



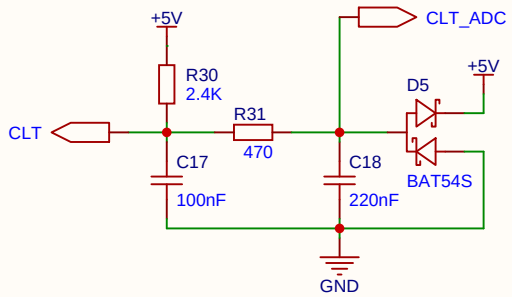
Schematic	ZTech ECU v0.1	Create at	2026-10-27
		Update at	2026-11-01
Board	Board1	Page	Power Management
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 6 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

Fuel Pump Driver



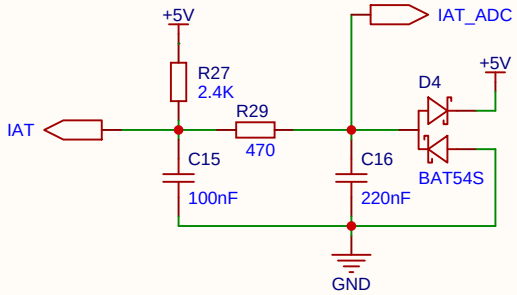
Schematic	ZTech ECU v0.1	Create at	2026-10-30
		Update at	2026-11-01
Board	Board1	Page	Fuel Pump
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version / Size	Page 7 Total 16
EasyEDA		V1.0 / A4	EasyEDA.com

COOLANT TEMP SENSOR



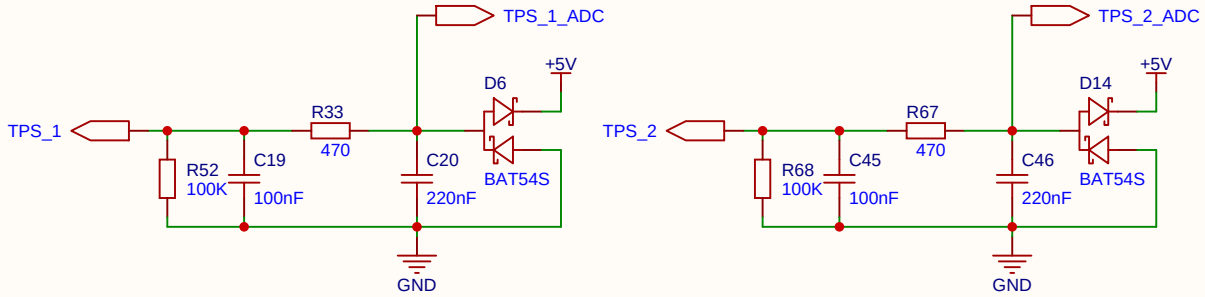
Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	CLT Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version / Size	Page 8 Total 16
EasyEDA		V1.0 / A4	EasyEDA.com

INLET AIR TEMP SENSOR



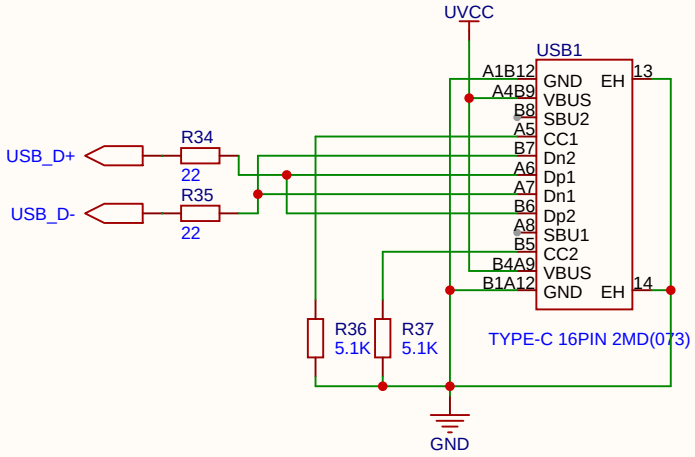
Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	IAT Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 9 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

THROTTLE POSITION SENSOR

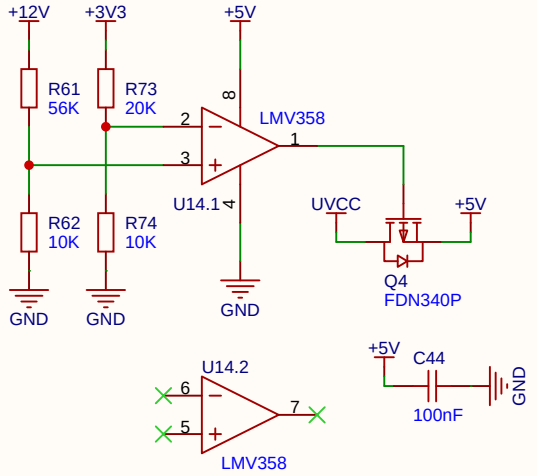


Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	TPS Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 10 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

USB-C CONNECTOR

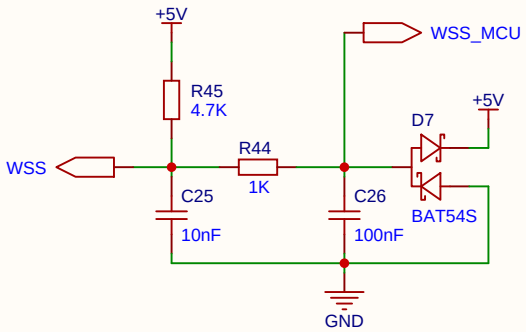


Digital Switch USB / BAT



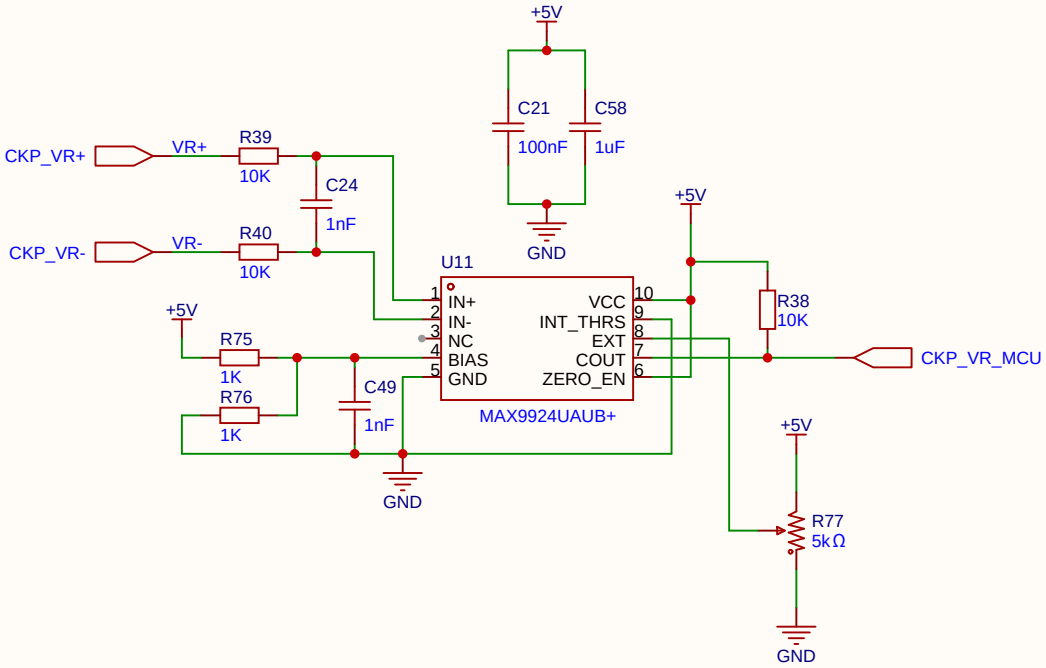
Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	USB
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 11 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

Speed Hall SENSOR

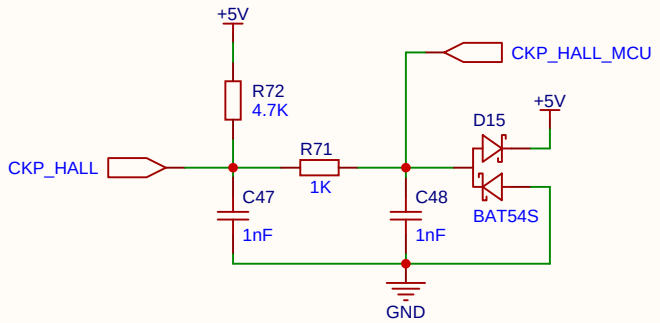


Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	Speed Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 12 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

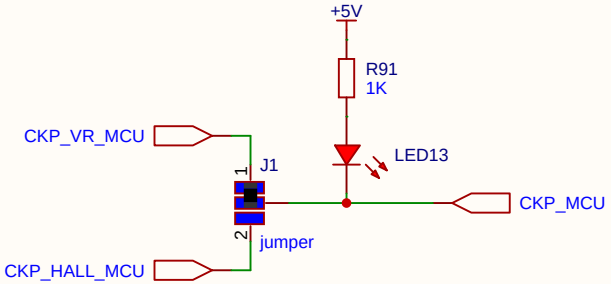
CKP VR SENSOR



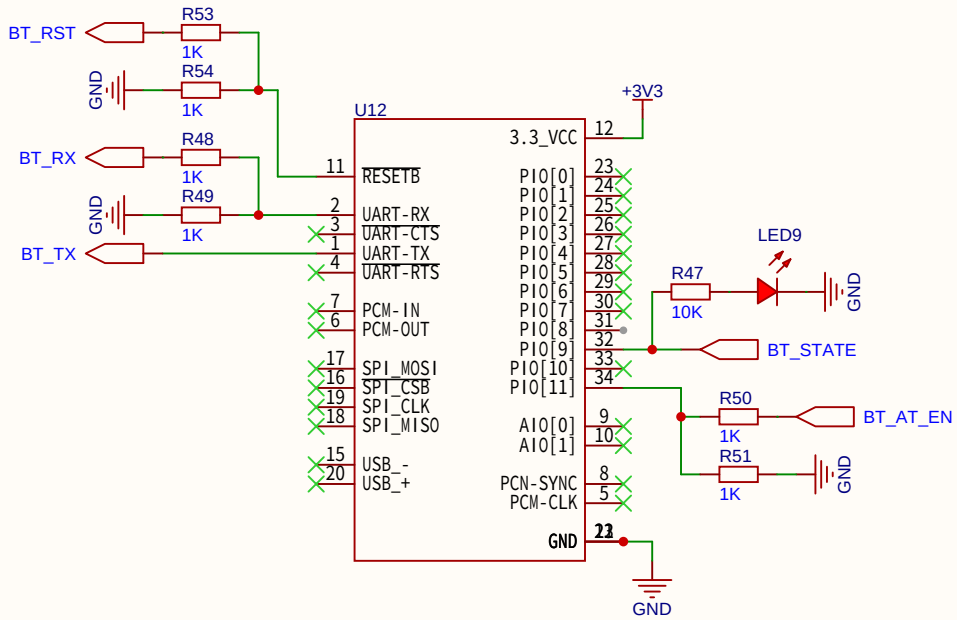
CKP HALL SENSOR



Jumper VR/HALL

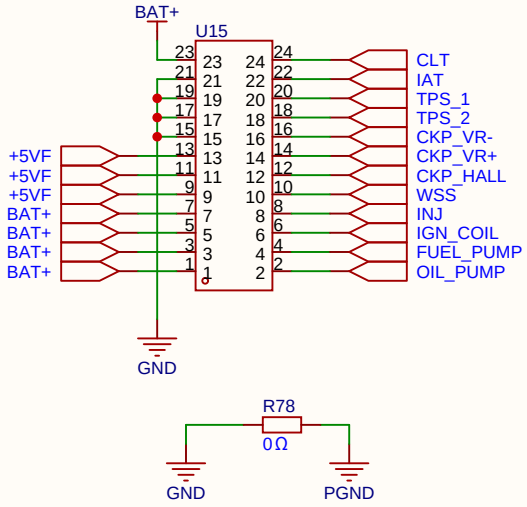


Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-02
Board	Board1	Page	CKP Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 13 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

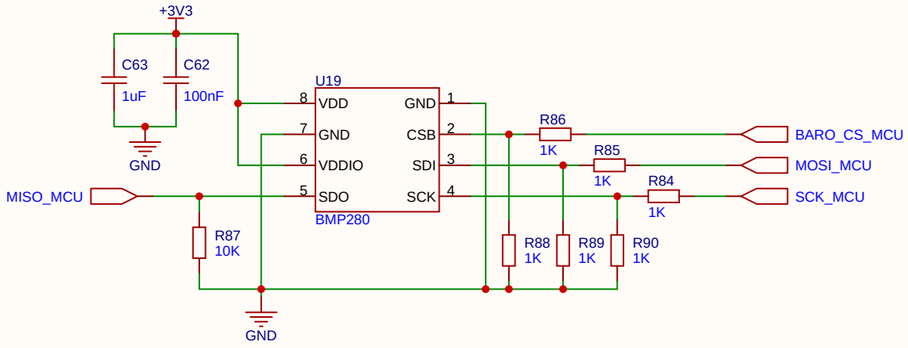


Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-01
Board	Board1	Page	Bluetooth
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 14 Total 16
EasyEDA		V1.0	A4	EasyEDA.com

24 PIN CONNECTOR



Schematic	ZTech ECU v0.1	Create at	2026-10-29
		Update at	2026-11-02
Board	Board1	Page	Connector
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 15 Total 16
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	ZTech ECU v0.1	Create at	2026-11-02
		Update at	2026-11-02
Board	Board1	Page	Barometer Sensor
Drawn		ECU_ATMEGA32U4_01.11.2026	
Reviewed			
		Version	Size	Page 16 Total 16
EasyEDA		V1.0	A4	EasyEDA.com